LABORATORY REPORT

COURSE CODE: CIP-B101

COURSE TITLE: Basic Computer Skills for Digital Forensics

LAB CODE: Lab 1B

LAB TITLE: PC Fundamentals, Disk Sectors, and Partition Analysis

Institution: International Cybersecurity and Digital Forensics Academy (ICDFA)

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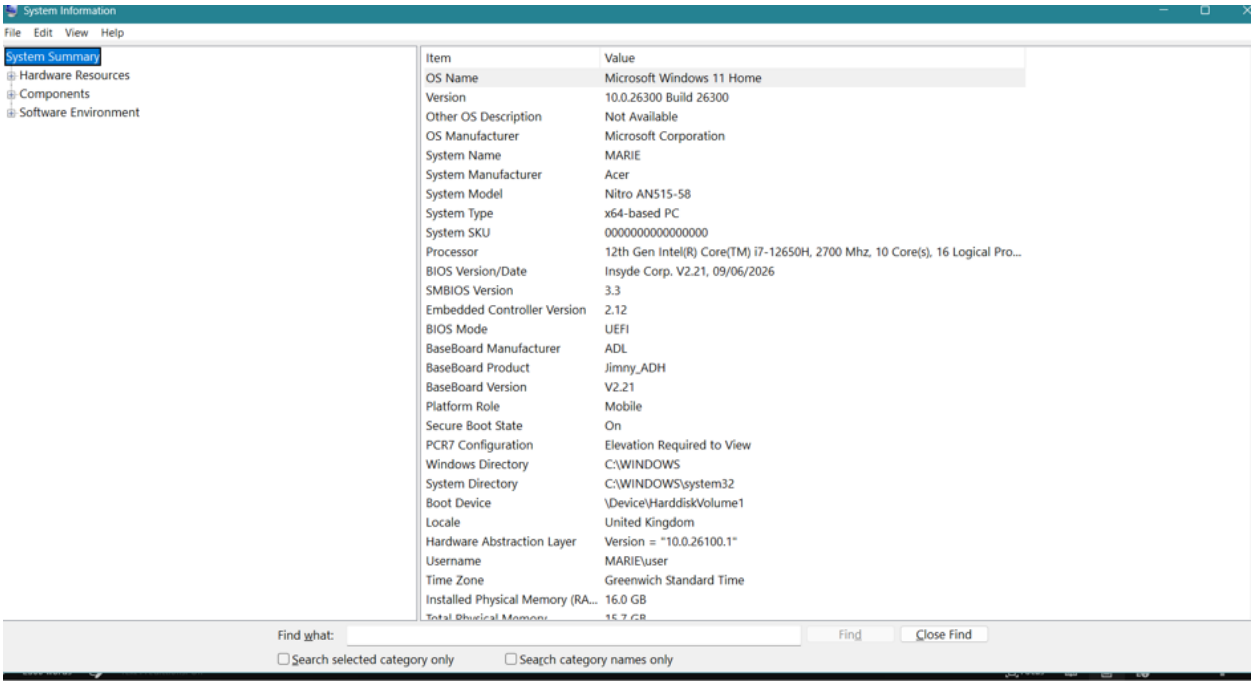
Student ID: C11/26/PCDFIR/17028

Platform used: Windows 11/Kali GNU/Linux 2026.2

INTRODUCTION

This laboratory exercise documents the hardware, software, memory, disk sector, and file system architecture of an investigation workstation using safe, read-only techniques. Digital forensics relies on maintaining data integrity, altering system stage during triage can invalidate digital evidence.

A1. Capture basic system specification



run the command below in Command Prompt and capture the first page of output.

```
Microsoft Windows [Version 10.0.26300.9711]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>systeminfo

Host Name:                     MARIE
OS Name:                       Microsoft Windows 11 Home
OS Version:                    10.0.26300 N/A Build 26300
OS Manufacturer:               Microsoft Corporation
OS Configuration:              Standalone Workstation
OS Build Type:                 Multiprocessor Free
Registered Owner:              user
Registered Organization:       N/A
Product ID:                    00342-22187-95362-AAOEM
Original Install Date:         01/11/2025, 04:40:29
System Boot Time:              26/10/2026, 05:37:05
System Manufacturer:           Acer
System Model:                  Nitro AN515-58
System Type:                   x64-based PC
Processor(s):                  1 Processor(s) Installed.
                               [01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2700 Mhz
BIOS Version:                  Insyde Corp. V2.21, 09/06/2026
Windows Directory:             C:\WINDOWS
System Directory:              C:\WINDOWS\system32
Boot Device:                   \Device\HarddiskVolume1
System Locale:                 en-gb;English (United Kingdom)
Input Locale:                  en-gb;English (United Kingdom)
Time Zone:                     (UTC+00:00) Monrovia, Reykjavik
Total Physical Memory:         16,088 MB
Available Physical Memory:     5,784 MB
Virtual Memory: Max Size:      17,424 MB
Virtual Memory: Available:     3,935 MB
Virtual Memory: In Use:        13,489 MB
Page File Location(s):         C:\pagefile.sys
Domain:                        WORKGROUP
Logon Server:                  \\MARIE
Hotfix(s):                     5 Hotfix(s) Installed.
                               [01]: KB5127694
                               [02]: KB5054156
                               [03]: KB5121794
                               [04]: KB5127648
```

A2. Capture storage information

```
Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\System32> Get-Partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_micron_3400_mtfd#4&3795839e&0&020000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    260 MB System
2                            273678336                                   16 MB Reserved
3                C           290455552                                952.52 GB Basic
4                            1023055757312                              1.07 GB Recovery


PS C:\WINDOWS\System32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
C           Acer         NTFS           Fixed     Healthy      OK                    626.37 GB 952.52 GB
                         NTFS           Fixed     Healthy      OK                    114.68 MB   1.07 GB


PS C:\WINDOWS\System32>
```

A3. Map evidence sources to system layers

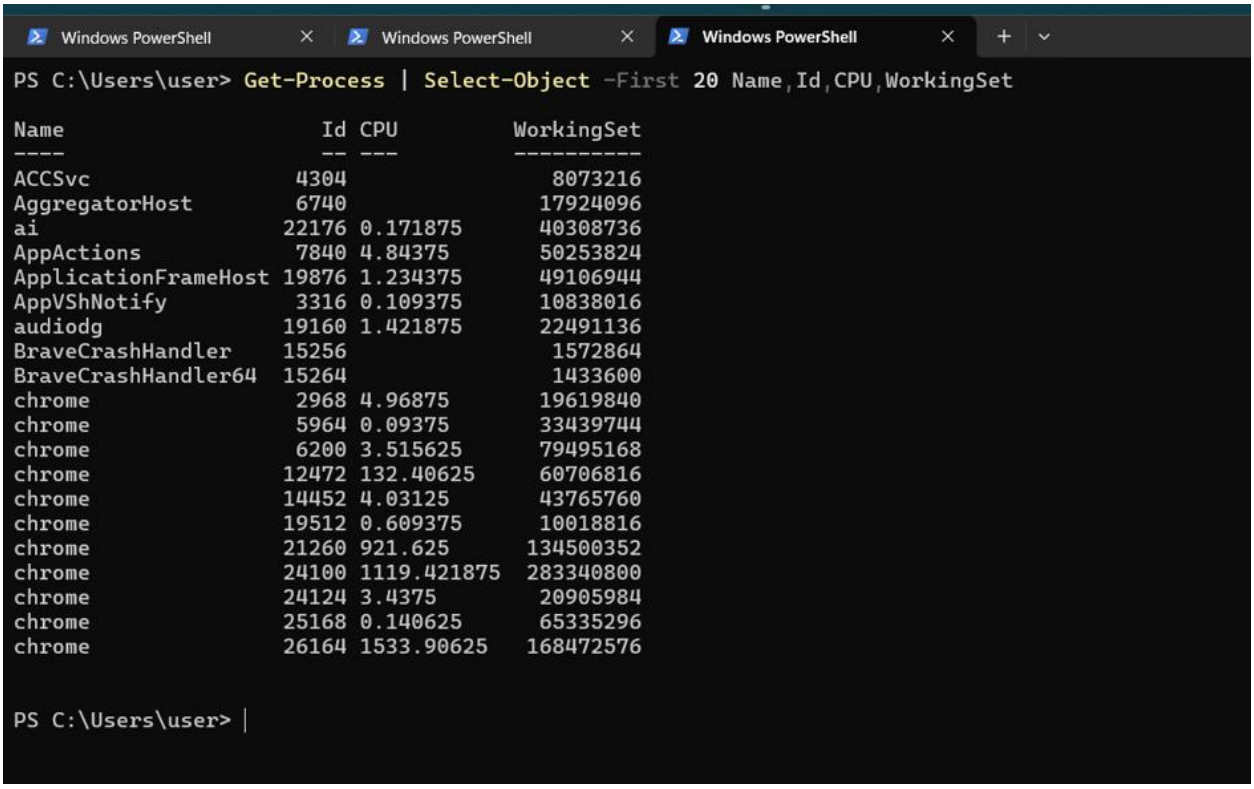
Layer / Source	Example Evidence	Where It May Be Found	Forensic Importance
User Layer	Shell history, recent document access, search terms, credential entries.	web browser profiles.	Links a specific human operator/account to malicious or unauthorized activity.
Application Layer	Office document metadata, chat databases, browser caches, SQLite DBs.	App-specific folders browser cache, and directories.	It demonstrates which application interacted with a specific artifact.
OS Layer	Event logs, Registry hives, audit logs, service installation logs.	Event Logs	Records OS-level system activity, user logons, and execution history.

Disk Layer	Master File Table (MFT), unallocated space, file slack, deleted fragments.	Physical sectors, file system structures, partition tables.	Allows recovery of deleted files, raw carving, and timeline reconstruction.
Memory / Process Layer	Injected code, volatile network sockets, running process tree.	Volatile RAM, crash dumps.	Captures live system state, and unencrypted credentials.

Explain why evidence may exist in more than one layer.

Because evidence frequently spans multiple system layers due to how data flows through computer architecture.

B1. Observe running processes



B2. Explain the difference

Explain the difference between a program, a process, and a thread in your own words.

Program: It is code that is stored on a computer in the form of a file

Process: It refers to a program in execution that has been allocated system resources such as memory, CPU time, etc.

Thread: This is the smallest unit of execution; there can be multiple threads in a single process.

C1. Reproduce the slide calculation

Use the values from the slide to calculate the total disk size: sectors per track = 400, heads = 12, cylinders = 17000, and bytes per sector = 512.

TotalSectors = 400 x 12 x 17,000 = 81,600.000 sectors

Total Bytes = 81,600,000 x 512 = 41,779,200,000 bytes.

Kilobytes = 41,779,200,000 ÷ 1,024 = 40,800,00 KB

Megabyte = 40,800,000 ÷ 1024 = 39,843.75 MB

Gi9gabytes = 39,843.75 ÷ 1,024 = 38. 909912109375GB or 38.91GB

Item	Formula / Value	Answer
Total sectors	400 x 12 x 17000	81,600,000 sectors
Total bytes	81,600,000 x 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 / 1024	40,800,000 KB
Megabytes	40,800,000 / 1024	39,843.75 MB
Gigabytes	39,843.75 / 1024	38.91 GB

C2. Calculate your own disk or partition size

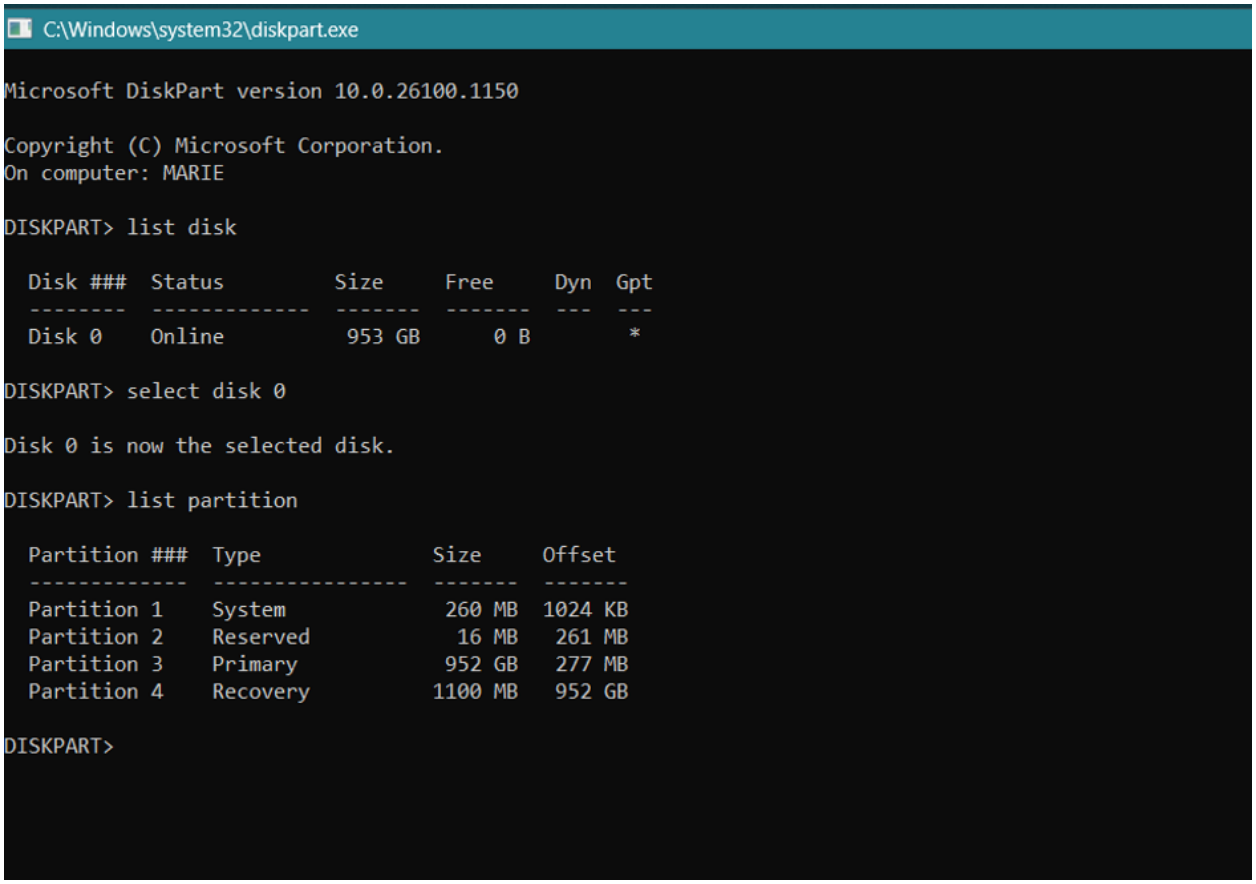
- Use your partition table or disk details to identify at least two partitions.
- For each partition, record the start sector, number of sectors, sector size, and file system where available.
- Calculate the size of each partition in GB using the formula above.
- Your calculation must be shown step-by-step. Screenshots alone are not enough.

STEP-BY STEP CALCULATION

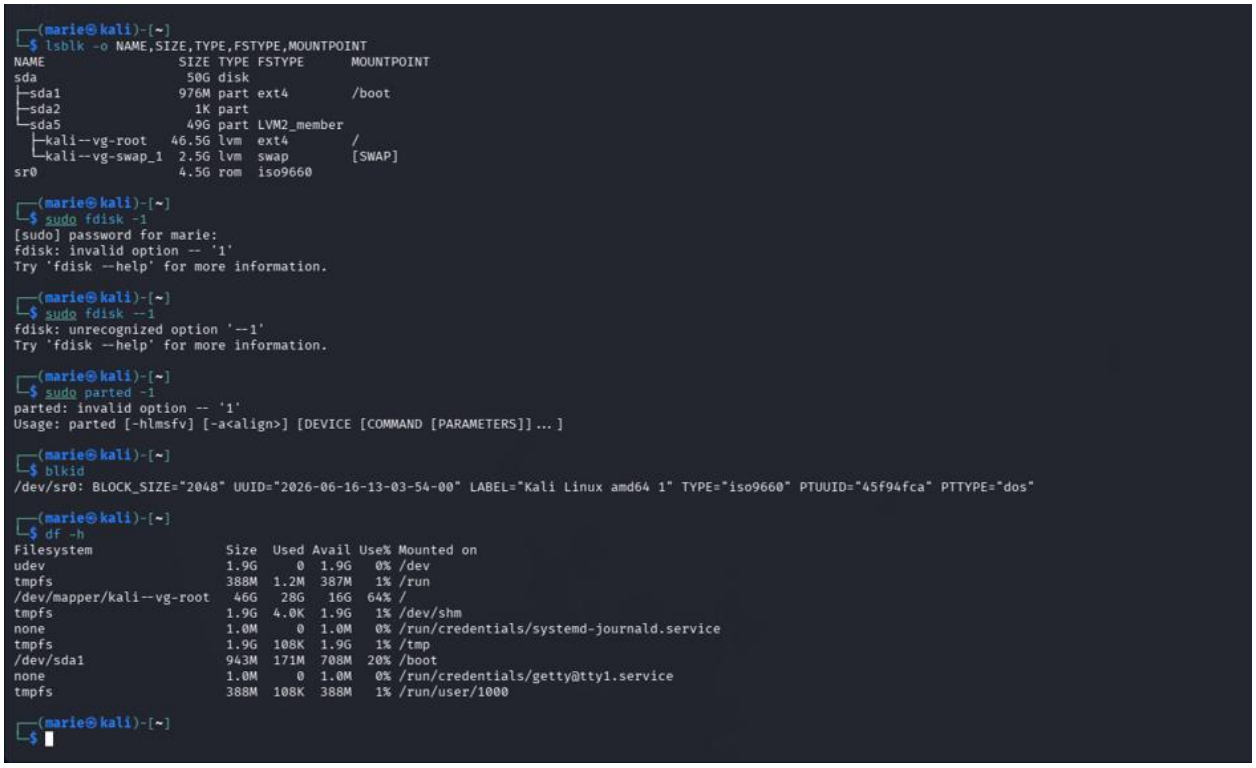
2,000,397,735 x 512 = 1,024,203.640,320 bytes.

Item	Value
Description	Disk drive
Manufacturer	(Standard disk drives)
Model	NVMe Micron_3400_MTFDKBA1T0TFH
Bytes/Sector	512
Media Loaded	Yes
Media Type	Fixed hard disk
Partitions	3
SCSI Bus	2
SCSI Logical Unit	0
SCSI Port	0
SCSI Target ID	0
Sectors/Track	63
Size	953.86 GB (1,024,203,640,320 bytes)
Total Cylinders	124,519
Total Sectors	2,000,397,735
Total Tracks	31,752,345
Tracks/Cylinder	255
Partition	Disk #0, Partition #0
Partition Size	260.00 MB (272,629,760 bytes)
Partition Starting Offset	1,048,576 bytes
Partition	Disk #0, Partition #1
Partition Size	952.52 GB (1,022,765,301,760 bytes)
Partition Starting Offset	290,455,552 bytes
Partition	Disk #0, Partition #2
Partition Size	1.07 GB (1,153,433,600 bytes)
Partition Starting Offset	1,023,055,757,312 bytes

D1 Windows method - Disk Management, PowerShell and DiskPart



D2. Linux/Kali method - read-only inspection



E1. Explain the boot process

1. Power-on : Motherboard firmware initializes hardware components such as the CPU and performs Power On Self-Test.
2. BIOS/UEFI: System reads firmware settings from CMOS and initializes boot sequence selection.
3. MBR / GPT Boot Record: Firmware accesses sector or checks GUID partition Table and reads EFI System Partition.
4. Boot Loader: Boot Manager loads into memory and parses boot configuration databases.
5. Kernel and OS Control: Operating system kernel loads system drivers, starts initial processes, and prompts user logon.

E2. Compare MBR and GPT

FEATURES	MBR	GPT
Maximum Disk Size	**2 Terabytes**	**9.4 Zettabytes**
Maximum Partition	**4 maximum partitions**	**128 partitions by default**
Boot Architecture	**Traditional BIOS**	**UEFI**

OS Support	**All operating systems, including legacy systems**	**Modern operating systems(Windows8/10/11)**

PART F

Explain why a file system is important in digital forensics.

The file system is important in digital forensics because it acts as a map for all data on a drive. It helps digital forensics locate evidence and recover deleted files.

CONCLUSION

This lab demonstrated basic computer architecture analysis without violating forensic data integrity. Using built – in diagnostics commands such as msinfo32, systeminfo command system configurations were documented in a non-destructive manner.

KEY TAKE AWAYS:

1. Digital evidence leaves artifacts across all systems; multi-layer correlation is required to establish proof in digital investigations.
2. Understanding the boundary between static programs, executing processes, and running threads is essential for live memory triage and detecting evasion techniques like process injections.
3. Calculating Disk sizes directly from raw sector counts ensures reliance on hardware truths rather than Operating System abstraction layer reporting.
4. Modern UEFI/GPT systems offer redundancy and capacity advantages over legacy BIOS/MBR systems. Partition tables direct how raw disk space is segmented and interpreted by forensic toolsets.